VINE: Variable Input N-view Encoder for 3D Generation from Partial Posed Observations

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Abstract

Image-to-3D models produce compelling shapes from a single image but hallucinate unobserved regions. When an object is captured from several viewpoints with partial observability, these hallucinations contradict the other views. Extending such models from single- to multi-view input is non-trivial: the number of views varies per capture, and partial observations cover disjoint regions, for which naive aggregation is insufficient to compose complementary evidence. We present VINE, a lightweight multi-view extension of a two-stage rectified-flow model for structured 3D latent generation. VINE introduces a multi-view encoder that processes an arbitrary number of posed images, encoding the full camera frustum as relative positional biases for geometrically consistent cross-view aggregation. The generation stages receive deliberately different conditioning: geometry relies on cross-view exchange to resolve shape from partial evidence, while texture preserves per-view detail and gains little from such exchange. We adapt the frozen denoiser with LoRA and a curriculum that gradually increases the number of input views, training on a single H100 with 16K instances. Because input poses are supplied by the capture system, VINE conditions on them directly rather than estimating them from sparse cross-view correspondence. Under a partial-observation evaluation protocol on Toys4K and OmniObject3D, VINE surpasses tuning-free single-view TRELLIS adaptations. Our model matches depth-conditioned methods on geometry while significantly improving rendered texture, using posed images alone, and extends to sparse-view scene reconstruction, where it significantly improves geometry over depth-conditioned baselines on ScanNet++.

1. Introduction

Recent image-to-3D models reconstruct convincing object shapes from a single image, raising the prospect of turning ordinary captures, such as a short video, into 3D assets. Yet there is a mismatch between how these models are built and how objects are actually captured. They are trained on centered renderings spanning the full viewing sphere, and they reconstruct unseen regions by hallucinating from a learned object prior. At the same time, during real captures, an object is observed only partially - no single view reveals most of its surface, and successive views cover different, often disjoint regions. This partiality breaks the single-image paradigm in two ways. First, there is no longer a "good" view to condition on, since every view shows only a fraction of the object; thus, any specific choice discards the evidence in the others. Second, the regions a partial view leaves unseen cannot be predicted from a generic prior alone, since they are constrained by the other views the user actually captured. The desired capability for partial captures, therefore, lies not in generation from an object prior but in the fusion of complementary partial evidence.

The natural remedy is to condition a model on multiple views. Yet existing multi-view approaches rest on assumptions that are violated by partial captures. Methods that condition on a fixed set of canonical views [36] require an object to be recorded in a specific way with a prescribed view count, which is not always possible in real captures. Tuning-free adaptations that reuse a single-view model by sampling or averaging across views [2, 34] avoid retraining, but collapse when views do not share their content. Methods that supplement generation with reconstruction features [1] depend on co-visible overlap to establish correspondence, which sparse partial views do not provide. Common to all is that none is trained to compose complementary observations, a capability that the partial regime demands.

Moreover, the existing image-to-3D evaluation protocols do not address the partiality concern. Standard benchmarks condition on full-coverage views, where the conditioning signal together with the object prior already determines the shape; under these conditions, a model that hallucinates well is indistinguishable from one that reconstructs faithfully. We argue that the partial-observation regime is both practically relevant and missing from existing protocols,

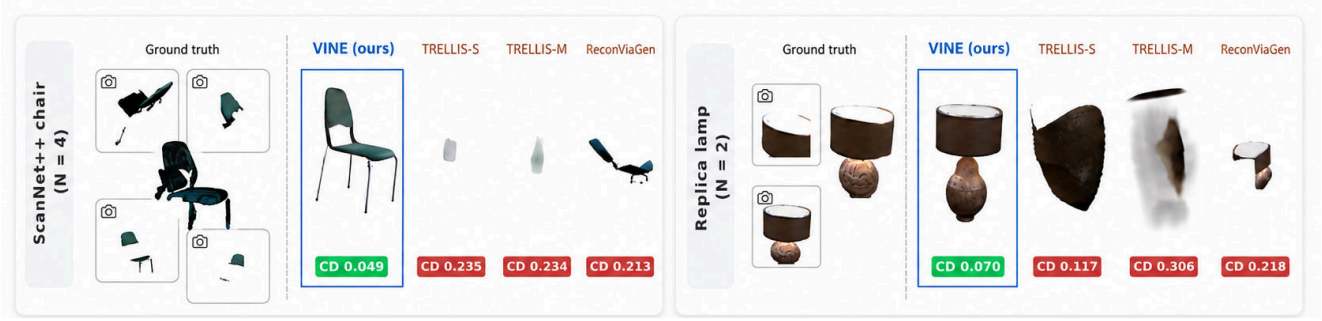


Figure 1. Given N partial posed captures of an object, VINE fuses them into a single shape consistent with all inputs. Tuning-free multi-view adaptations of the same frozen TRELLIS backbone (TRELLIS-S/M) collapse when conditioning views cover disjoint regions, and the VGGT-conditioned ReconViaGen degrades without sufficiently distinctive co-visible overlap.

and we introduce a benchmark that simulates it directly.

With that, we propose a model that leverages multi-view image conditions to faithfully generate object shapes from an arbitrary number of partial captures. We design a multi-view encoder with pose injection that provides a multi-view condition to the model, surpassing single-view performance once multiple views are available and enabling the generation of shapes that are more faithful to all input views. We further show that the two generations require opposite forms of multi-view conditioning and design the encoder accordingly. Lastly, we demonstrate the performance of our model on the novel partial-view benchmark and its application in real-world scene reconstruction, as shown in Fig. 1. Our contributions can be summarized as follows:

- We propose a lightweight multi-view encoder with pose injection that aligns a single-view generation model towards a multi-view posed input. We design a training strategy for a two-stage generative model demonstrating asymmetry in the alignment of geometry and texture.
- We propose a partial-observation evaluation protocol on 3D datasets [24, 31] exposing the fusion-vs-hallucination distinction, which standard full-coverage benchmarks cannot, and show that VINE achieves statistically significant gains in rendered texture quality on both benchmarks, with geometry that is significantly better from $N \geq 2$ on OmniObject3D.
- Our evaluation on indoor-scene datasets [25, 37] from [21] further tests the applicability of our model on real-world sparse reconstruction, with statistically significant geometry gains over all depth-conditioned baselines on ScanNet++ and over the SAM3D baselines on Replica.

2. Related Work

Multi-view 3D Generation. Image-conditioned 3D generation began with [15], synthesizing novel views of an object from a reference image and a target relative camera pose. Addressing the multi-view inconsistency of single-image models [14, 17, 18], later methods directly produce latent representations in a single forward pass [33–36].

Conditioning on multiple images opens a richer input space. ReconViaGen [1] supplements generation with VGGT [26] geometric features, though this requires co-visible overlap to establish cross-view correspondences, while Hunyuan3D 2.0 [36] extends to multi-image conditioning but restricts inputs to fixed canonical views. Tuning-free adaptations of single-view models [2, 34] avoid retraining by conditioning on a different view per denoising step (stochastic) or averaging predicted velocities across views (multidiffusion), and MV-SAM3D [10] replaces naive averaging with visibility- and entropy-aware weighting. Concurrently, Pixal3D [11] back-projects pixels into a 3D feature volume aggregated across views. All of the above assume substantial co-visible surface area, which breaks in partial-view settings, where observations cover disjoint regions and no single view reveals most of an object's geometry. VINE instead conditions a pretrained single-image model on arbitrary relative camera poses via PRoPE, paired with a curriculum that trains it to aggregate evidence from non-overlapping observations.

Object Reconstruction in the Wild. In real recordings, objects are rarely fully visible due to occlusion or field-of-view truncation. Amodal3R [32] conditions TRELLIS on an occlusion mask to guide completion, but such masks are rarely available without manual annotation. ShapeR [23] recovers unseen geometry from rich multi-modal conditioning (SLAM-derived point clouds, camera poses, text, and images), at the cost of 3D-sensor access and per-object annotation, while SAM3D [2] relies on an extensive multi-stage pipeline for occlusion robustness. Scene-level methods such as SceneGen [19] reconstruct full rooms in a single pass but lack per-instance control, yielding unreliable amodal completion at object boundaries. VINE instead needs only lightweight LoRA fine-tuning on a 16k-instance partial-view dataset to extend to real scenes with faithful, view-consistent completion.

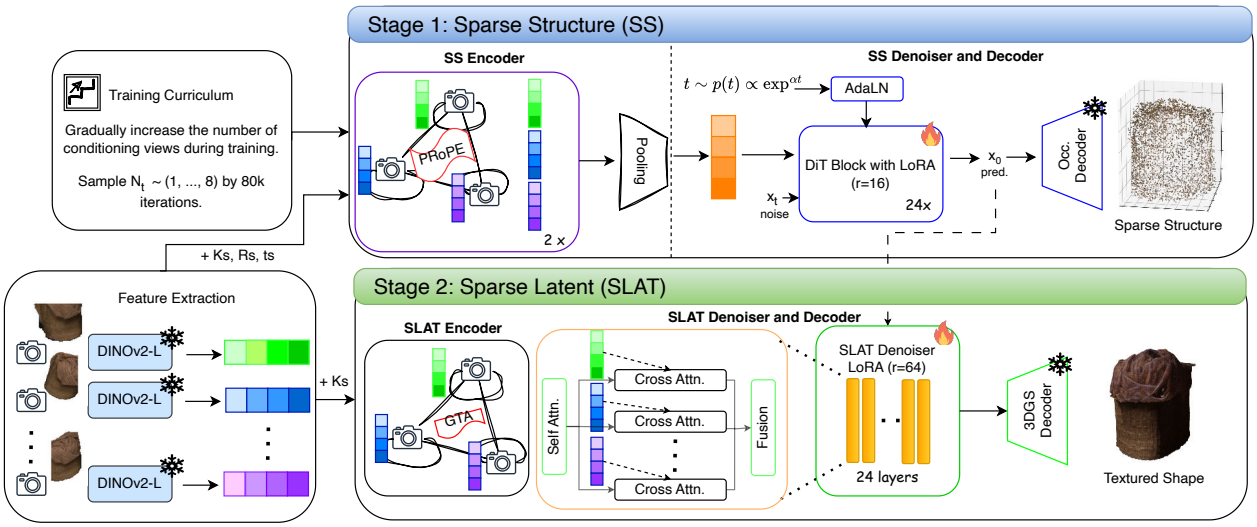


Figure 2. Given N posed images with object masks, the *SS encoder* interleaves PRoPE cross-view attention layers with per-frame self-attention passes to refine each view's tokens, then combines all N token sets into a single-view condition via attention pooling. The *SLAT encoder* ($L=1$) applies a single GTA layer for spatial alignment, returning a per-view list; each denoiser block cross-attends to every view independently, preserving view-specific texture cues. Both encoders attach to the frozen TRELLIS [34] denoisers via LoRA adapters. The SS stage is trained by gradually increasing the number of input views, whereas the SLAT stage benefits from random set selection.

3. Method

We extend TRELLIS [34], a single-image-to-3D generation model, to accept an arbitrary number of posed multi-view images as conditioning input. As shown in Fig. 2, our approach introduces a lightweight multi-view encoder that aggregates cross-view information using camera-aware attention (PRoPE [12]), and attaches to the frozen denoiser via low-rank adapters. We first review the relevant background (Sec. 3.1), then describe our architecture (Sec. 3.2) and training strategy (Sec. 3.3).

3.1. Background

TRELLIS [34] represents 3D objects as Sparse Structured LATents (SLAT), a sparse 3D grid, where each voxel carries a latent vector extracted by aggregating DINOv2 features from multi-view renderings of the object. SLAT can be decoded into multiple output representations (Radiance Fields, 3D Gaussians, or meshes) by plugging in interchangeable decoders, while the latent itself remains compact and representation-agnostic.

Generation proceeds in two sequential stages. In the Sparse Structure (SS) stage, a diffusion transformer denoises a binary sparse voxel occupancy field that defines the coarse geometry. In the Sparse Structure Latent (SLAT) stage, a second transformer adds texture to the sparse structure and fine geometric details. The forward process follows the rectified flow matching objective [13, 16] and linearly interpolates between clean data x_0 and noise ε: $x_t = (1-t)\,x_0 + t\varepsilon$ with $t \in [0,1]$. The primary loss term for both stages, denoted as u_θ, is the velocity mean-squared error given target velocity u^*, which moves noisy samples towards the true data distribution:

$$\mathcal{L}_u = \mathbb{E}_{t,\varepsilon}\left[\|u_\theta(x_t,t,c) - u^*\|^2\right], \tag{1}$$

where c is an encoded condition (text or an RGB image).

Projective Positional Encoding (PRoPE)[12] is a camera-aware positional encoding for transformers that encodes the full camera frustum, both intrinsics and extrinsics, as relative positional biases in attention. Rather than appending camera parameters as extra tokens or adding a separate conditioning branch, PRoPE encodes the full camera frustum as block-diagonal transforms applied to the queries, keys, values, and output of each attention layer. In comparison, GTA [20] encodes only relative SE(3) camera extrinsics as attention transforms.

Concretely, each camera i defines a full 4×4 projection matrix $\tilde{\mathbf{P}}_i = \widehat{\mathbf{K}}_i\,\mathbf{T}_i$, where $\widehat{\mathbf{K}}_i$ is the normalized intrinsics matrix and $\mathbf{T}_i \in SE(3)$ is the world-to-camera extrinsics. PRoPE applies block-diagonal transforms to queries, keys, values, and the attention output: queries by $\tilde{\mathbf{P}}_i^\top$, keys and values by $\tilde{\mathbf{P}}_i^{-1}$, and the output by $\tilde{\mathbf{P}}_i$. The resulting query-key inner product implicitly computes the relative projective transform $\tilde{\mathbf{P}}_i\tilde{\mathbf{P}}_j^{-1}$ for each pair of views (i,j), establishing patch-level geometric correspondences and enabling geometry-aware context exchange across views.

3.2. Multi-View Encoder with Pose Injection

Given N posed images and corresponding object masks $\{(\mathbf{I}_i, \mathbf{T}_i, \mathbf{K}_i)\}_{i=1}^N$, where $\mathbf{T}_i \in SE(3)$ is the world-to-

camera camera pose and $\mathbf{K}_i \in \mathbb{R}^{3\times3}$ is the intrinsic matrix, we follow TRELLIS [34] pipeline and extract per-view DINOv2 [22] features $\mathbf{F}_i \in \mathbb{R}^{P\times C}$, where P denotes the number of tokens and C - their feature dimension. The per-view features are stacked into a joint sequence $\mathbf{F} \in \mathbb{R}^{N\times P\times C}$.

Multi-view Encoder with PRoPE. We process the features with L transformer layers, each performing full attention over NP tokens with PRoPE camera conditioning as described in Sec. 3.1. Similar to [7], we apply PRoPE attention not only to patch tokens as advocated in [12], but also to global tokens to encourage cross-view global exchange.

Learnable view aggregation. When aggregation is enabled (Sec. 3.3), the encoder fuses N per-view feature sequences into a single sequence of P tokens, matching the conditioning interface of the frozen TRELLIS denoiser. We adopt attention pooling: at each patch position, the N per-view features attend to each other through a single self-attention layer, refining each view's representation with cross-view context, and are then aggregated by averaging over views.

3.3. Two-Stage Training Strategy

Geometry reconstruction and texture generation impose fundamentally different demands on multi-view conditioning: while sparse voxel occupancy prediction is a global inference problem, texture generation, by contrast, is view-dependent and local. A geometry denoiser must let all views exchange context with each other to resolve which voxels are occupied, but is indifferent to whether that context is later collapsed into a single conditioning signal. Texture generation, in contrast, needs each view's tokens to survive intact through the denoiser: premature aggregation blurs unique surface appearance cues, including view-dependent shading, while cross-view exchange contributes little, since appearance is already locally well-constrained per view. This contrast drives the two encoder and their integration designs.

SS stage. Inspired by [26], the Sparse Structure (SS) encoder ($L = 2$) interleaves PRoPE cross-view attention layers with per-frame self-attention passes to further enhance the expressiveness of each token set. The resulting features are then pooled across views via attention pooling into a single sequence of P tokens, forming a unified conditioning signal that captures the aggregate geometry evidence.

Moreover, the model must unlearn the single-view spatial ambiguities it has absorbed during pretraining and instead integrate consistent cross-view evidence. Randomly switching the number of multi-view inputs produces large initial gradient updates that interfere with the pretrained occupancy prior and makes the alignment job significantly harder. We therefore use a training curriculum: starting with single-pose-view conditioning and linearly increasing the view count to $N_{\max} = 8$ over 80K steps, sampling uniformly from $\{1, \ldots, N_t\}$ at each step. This slower adaptation stabilizes early training and lets the model build multi-view geometry reasoning incrementally.

SLAT stage. The Sparse structure LATent (SLAT) encoder ($L = 1$) uses a single GTA [20] layer, which encodes relative extrinsics only, rather than the full-frustum PRoPE [12] used in the SS stage. The encoder forgoes pooling entirely and instead returns a list of N per-view feature sequences. Each transformer block in the SLAT denoiser cross-attends to every token set, preserving the view-specific texture condition throughout the denoising process. Since texture appearance is inherently view-dependent, the model benefits most from exposure to the full diversity of view combinations from the outset. Imposing a curriculum would delay learning from multi-view texture configurations. We thus sample a random number of views at each step, giving the SLAT denoiser an immediate and balanced experience across all possible input counts.

4. Evaluation

We organize this section as follows: first, we outline the implementation details and the training protocol (Sec. 4.1). We proceed with the evaluation of our model on the object-centric benchmark (Sec. 4.2), where we simulate partial observability, and on the real, scene-level captures (Sec. 4.3). In Sec. 4.4 we report ablation studies validating our design.

4.1. Implementation details

Timestep Sampling. Rather than sampling timesteps uniformly, we bias them toward higher noise levels by drawing $t \in [0, 1]$ from a normalized exponential density

$$p(t) \propto e^{at}, \quad a = 4. \tag{2}$$

This concentrates probability mass near $t = 1$, where the denoiser has the most to learn, while the exponential tail still covers lower timesteps.

Lightweight encoder and efficient LoRA fine-tuning. VINE extends TRELLIS-image-large model [34], which has in total 1.2B parameters. Stage 1 (SS) parameters amount to 555.5M, while stage 2 (SLAT) parameters count reaches 600.4M. The multi-view encoder is lightweight (37.8M parameters in the SS stage and 12.6M parameters in the SLAT stage) and fully trainable. Rather than fine-tuning the entire denoiser, we inject Low-Rank Adaptation (LoRA) adapters [6] into all 24 attention blocks of each denoiser, keeping all original weights frozen. Combined with the multi-view encoder, this limits the total number of trainable parameters to 7.2% of the SS denoiser and 5.4% of the SLAT denoiser (42.9M and 33M trainable parameters, respectively), enabling training on a single GPU. We provide the exact per-component breakdown in the supplementary.

Hyperparameters. Both stages use AdamW (lr $= 10^{-4}$, cosine decay) and classifier-free guidance with 10% unconditional probability on 16K instances from Objaverse [4],

Method	$N=1$ views			$N=2$ views			$N=4$ views			$N=8$ views			Mean		
	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR
Amodal3R [32]	5.37	0.350	12.09	4.63	0.401	11.89	4.87	0.397	11.94	4.36	0.409	11.95	4.81	0.389	11.97
ReconViaGen [1]	8.48	0.240	15.70	8.10	0.261	15.92	8.55	0.257	15.67	9.00	0.243	15.58	8.53	0.250	15.72
TRELLIS-S [34]	7.08	0.309	15.87	6.29	0.334	16.45	6.06	0.365	17.04	6.08	0.369	17.02	6.38	0.344	16.60
TRELLIS-M [34]	7.08	0.309	15.87	5.97	0.355	16.62	6.12	0.355	17.16	5.89	0.374	17.49	6.27	0.348	16.79
SAM3D-M [2]	4.13	0.463	17.54	3.28	0.517	18.36	2.75	0.589	19.14	2.81	0.583	19.22	3.24	0.538	18.56
SAM3D-S [2]	4.15	0.463	17.56	3.29	0.520	18.33	2.72	0.595	19.13	2.80	0.588	19.21	3.24	0.542	18.56
MV-SAM3D [10]	4.17	0.466	17.49	3.44	0.518	18.31	2.74	**0.602**	19.20	2.91	0.586	19.14	3.32	0.543	18.54
Ours	**3.81**	**0.486**	**17.87**	**3.05**	**0.547**	**18.64**	**2.59**	0.599	**19.69**	**2.30**	**0.627**	**20.18**	**2.94**	**0.565**	**19.10**

Table 1. Quantitative comparison on the partial-view OmniObject3D benchmark (200 instances), plus the per-method average across $N \in \{1, 2, 4, 8\}$. CD×100 (↓), F@.02 (↑), PSNR in dB (↑). **Bold**: best; underline: second best per column. Tuning-free variants (S - stochastic, M - multidiffusion) of TRELLIS [34] and SAM3D [2] are evaluated. VINE achieves the best scores for CD and PSNR at every view count, while the former improves monotonically from $N=1$ to $N=8$.

ABO [3], and HSSD [9], with images at 518×518 resolution (dataset distribution details are in supp.). Beyond this shared setup, the two stages differ in several respects. The SS stage uses LoRA rank $r = 16$ and AdamW weight decay 0.01. The SLAT stage uses a higher LoRA rank $r = 64$ to accommodate the richer texture conditioning, sets weight decay to 0, and additionally receives the SS stage output as a geometric prior [34].

4.2. Multi-view Generation

Datasets. We evaluate the generalization capability of our model on two unseen object-centric datasets: Toys4k [24] and OmniObject3D [31]. Each test set contains 200 objects, with every instance providing multi-view posed images and a ground-truth point cloud from a voxelized mesh. At test time, we condition on $N \in \{1, 2, 4, 8\}$ input views selected from a 24-camera pool that uniformly covers the unit sphere, and evaluate reconstruction quality on a random subset of 10 held-out views. For every N, including $N=1$, views are drawn from the same 24-view candidate pool and selected to minimize a combined face-visibility, incidence, and pixel-density coverage score, yielding the most challenging subset available at that view count. Further details on the benchmark creation are provided in the supplementary material.

Protocol validation. We validate the partial protocol in two ways. Firstly, we perform a non-circularity check by projecting each object's point cloud and measuring in-frame silhouette fraction, both per-view (mean) and aggregated across a subset (union), shown in Tab. 3. Individual partial views are genuinely incomplete (72–87% in-frame, vs. 100% for a full-coverage control built on the same instances), yet the union across a subset is highly complementary, reaching 99.9–100% by $N = 4$–8. This validates that the selected views are individually hard but collectively cover the object. Secondly, we directly test whether the protocol is non-trivial. To this end, we check whether per-method rankings remain unchanged between the partial protocol and a standard control, such as a full-coverage or a clustered one. Tab. 3 reports the rank correlation between the proposed partial protocol and two controls built on the same instances, namely *full-coverage* (evenly spaced views) and *clustered* (intentionally redundant views). Partial vs. full-coverage correlation is negative at every N, and the best-CD method changes at every N, confirming that the protocol is not a relabeling of a standard benchmark, but a different test setting.

Metrics. We report three complementary metrics: Chamfer Distance (CD×100, ↓) between predicted and ground-truth point clouds; F-metric (F@.02, ↑), measuring precision and recall jointly; and PSNR (dB, ↑) on novel-view renderings. Predicted shapes are aligned to GT via conventional single-start ICP before evaluation, without the multi-start search or scale alignment used at scene level. All metrics are computed in the normalized object coordinate frame. We report paired Wilcoxon statistical significance tests [30] in the supplementary.

Baselines. We compare against TRELLIS [34] in stochastic (TRELLIS-S) and multi-diffusion (TRELLIS-M) adaptation modes; analogous variants of SAM3D - stochastic (SAM3D-S) and multidiffusion (SAM3D-M); MV-SAM3D [10], ReconViaGen [1], and the stochastic multi-view adaptation of Amodal3R [32]. Amodal3R and ReconViaGen, like VINE, build on TRELLIS [34], enabling a backbone-matched comparison that isolates the contribution of our multi-view conditioning. Additional comparison against Hunuyan3D-2mv [36] and Shape-R [23] is provided in the supplementary. All baselines are evaluated under the same ICP alignment. We do not compare against feed-forward reconstruction methods such as VGGT [26], as they recover only the visible geometry and cannot complete unseen surface regions, which is the core challenge of our benchmark.

Results on OmniObject3D and Toys4k are reported in 1 and Tables 2 respectively. VINE attains the best average scores on OmniObject3D and remains highly competitive on Toys4k. The most consistent signal is rendered texture quality: on OmniObject3D, paired tests (see supplementary) show VINE significantly improves PSNR over MV-SAM3D and SAM3D-M at every view count, while

Method	$N=1$ views			$N=2$ views			$N=4$ views			$N=8$ views			Mean		
	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR
Amodal3R [32]	5.41	0.355	12.32	4.70	0.408	12.36	4.64	0.427	12.45	4.19	0.443	12.45	4.74	0.408	12.40
ReconViaGen [1]	7.98	0.286	15.93	6.63	0.334	16.65	6.82	0.320	16.65	6.99	0.313	16.46	7.11	0.313	16.42
TRELLIS-S [34]	6.48	0.354	15.99	6.04	0.381	16.71	5.29	0.431	17.28	5.20	0.421	17.46	5.75	0.397	16.86
TRELLIS-M [34]	6.49	0.354	15.98	5.66	0.386	16.73	5.03	0.444	17.63	4.63	0.469	18.15	5.45	0.413	17.12
SAM3D-M [2]	**4.20**	0.435	16.96	3.25	0.513	17.95	**2.45**	0.603	18.94	2.35	0.623	19.28	**3.06**	0.544	18.28
SAM3D-S [2]	4.23	**0.435**	16.94	3.19	0.522	17.91	2.48	0.594	18.84	2.37	0.614	19.13	3.07	0.541	18.21
MV-SAM3D [10]	4.51	0.427	16.79	3.48	0.512	17.68	2.73	0.610	18.81	2.78	0.628	19.15	3.38	0.544	18.11
Ours	4.42	0.429	**16.98**	**3.11**	**0.541**	**18.09**	2.59	**0.613**	**19.05**	**2.23**	**0.670**	**19.68**	3.09	**0.563**	**18.45**

Table 2. Quantitative comparison on the partial-view Toys4k benchmark (200 instances) evaluated across different numbers of condition views, plus the per-method average across $N \in \{1, 2, 4, 8\}$. CD×100 (↓), F@.02 (↑), PSNR in dB (↑). **Bold**: best; underline: second best per column. VINE achieves the best average F@.02 and PSNR, with strong PSNR gains over depth-aware SAM3D tuning-free variants, while maintaining competitive average CD.

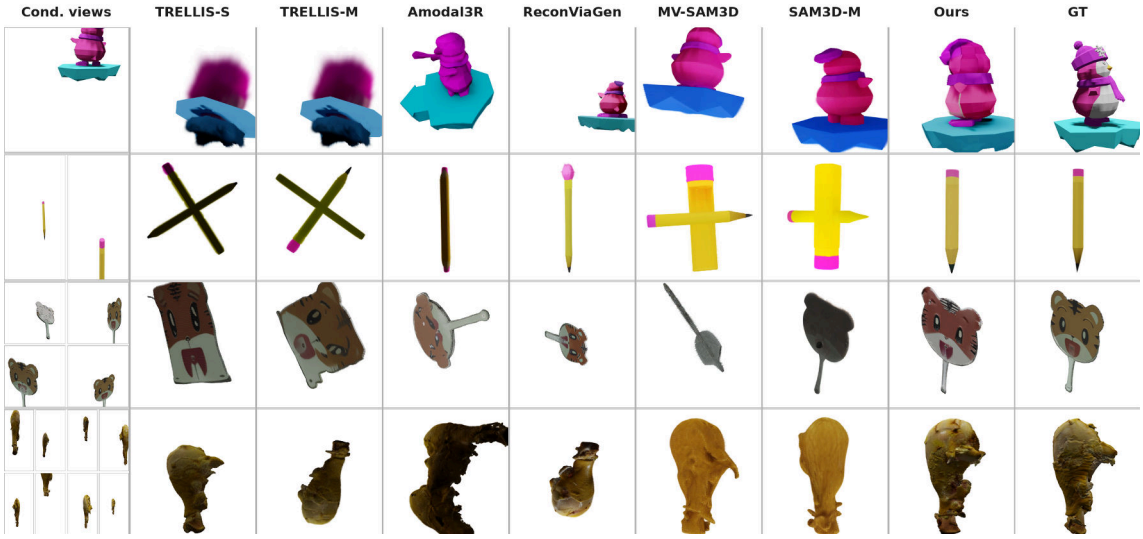


Figure 3. Qualitative comparison across $N = 1$–8 input views on Toys4k and OmniObject3D. All generated shapes are GT-aligned with ICP. Each row uses a different view count to stress-test the methods on different partial views. Our method recovers correct geometry and remains consistent with the textures in all cases, whereas baselines struggle with occlusion ambiguity and the limitations of naive fusion.

N	in-frame silhouette			Part.–Full.		Part.–Clust.		Full.–Clust.	
	Mean	Union	Full.	ρ	τ	ρ	τ	ρ	τ
1	0.719	0.719	1.000	−0.10	−0.07	−0.14	−0.21	0.95	0.86
2	0.816	0.967	1.000	−0.81	−0.71	−0.43	−0.36	0.52	0.36
4	0.845	0.999	1.000	−0.62	−0.50	−0.24	−0.14	0.60	0.50
8	0.867	1.000	1.000	−0.50	−0.36	−0.29	−0.14	0.67	0.50

Table 3. **Left:** Object-in-frame silhouette fraction under the proposed partial protocol (per-view mean and union across the N-view subset) vs. the full-coverage protocol (i.e., 1.0 at every N). **Right:** Spearman ρ / Kendall τ rank correlation of per-method CD rankings between each pair of protocols. Results from 8 methods on the Toys4k 200 instances show an inverse correlation with both full-coverage and clustered protocols — a different testing regime.

on Toys4k, PSNR gains over these baselines reach significance only at $N = 8$. Geometry gains are less consistent: on OmniObject3D, VINE significantly improves CD over MV-SAM3D at $N \in \{1, 2, 8\}$ and F@.02 at $N \in \{2, 8\}$. On Toys4k, geometric differences relative to the closest SAM3D variants remain within paired-test noise. It is worth noting that SAM3D utilizes metric depth estimator [27] as a geometric model prior, whereas our model uses camera information alone without any explicit point cloud information. The closest dedicated multi-view baseline, MV-SAM3D, adds visibility-aware weighting yet still scores lower on F@.02 and PSNR on OmniObject3D. Both TRELLIS adaptations lag roughly a factor of two on CD, confirming that naive aggregation is insufficient, while ReconViaGen mainly suffers from scale inconsistency, as its VGGT [26] pose conditioning is unreliable under few cross-view correspondences. Amodal3R shows low image quality due to a multi-view inconsistent completion.

Where the gains originate. Decomposing the improvement reveals a second asymmetry, orthogonal to the design asymmetry of Sec. 4.4: not *how* we condition each stage, but *where* the gain originates. On Toys4k, a single-view model adapted to partial data (a LoRA baseline with single-image conditioning and no multi-view encoder, see supp.) already reaches CD×100 3.94 at $N = 1$, recovering 2.54 of the 4.25 CD gap between the un-adapted base model (6.48)

and VINE at $N = 8$ (2.23). This signals that most of the geometric improvement comes from partial-data *adaptation* alone, before any cross-view fusion. Multi-view *fusion* then contributes a further 1.71 CD, but the larger share goes to the texture gain — 2.27 of the 3.69 dB total PSNR improvement, against 1.42 dB from adaptation. Geometry is thus driven chiefly by adaptation to the partial-view distribution, while texture is driven by cross-view fusion. Consistent with the design asymmetry above, cross-view exchange is still needed to realize geometry's (smaller) fusion increment (Tab. 5), so geometry lands near parity while the fusion-dominated texture shows clear, significant gains.

Figure 3 shows qualitative comparisons across $N = 1$ to $N = 8$ input views on Toys4k and OmniObject3D instances, which demonstrate the partial nature of conditioning views. Our model is capable of correctly fusing geometric and textural information from a variable number of views, while other methods are biased by individual reconstructions generating suboptimal shapes or colors. We provide additional examples along with an analysis of the models' performance with an increasing number of conditions in the supplementary material.

	Replica		ScanNet++	
Method	CD×100	F@.05	CD×100	F@.05
MV-SAM3D* [10]	62.13	0.057	62.00	0.061
TRELLIS-M [34]	9.26	0.533	7.58	0.620
ReconViaGen [1]	6.36	0.604	5.89	0.627
SAM3D-M [2]	5.24	0.651	<u>4.62</u>	<u>0.676</u>
MV-SAM3D+ICP [10]	<u>4.79</u>	<u>0.689</u>	4.69	0.671
Ours	**4.40**	**0.710**	**3.89**	**0.695**

Table 4. Quantitative comparison on the DP-Recon benchmarks[21]. CD×100 (↓) and F-metric (↑). All methods use a multi-start ICP with isotropic scale alignment except MV-SAM3D* (predicted pose). **Bold**: best; <u>underline</u>: second best per benchmark. Our model attains the best CD and F-score on both datasets, with a significant CD margin over the depth-aware SAM3D variants on ScanNet++.

4.3. Object Reconstruction in Real Scenes

Datasets. We evaluate on two scene-level benchmarks (Replica [25] and ScanNet++ [37]) with ground-truth object meshes from DP-Recon [21]. Replica [25] benchmark comprises 8 indoor scans, each containing 10 RGB images for training and 10 for testing, camera parameters (extrinsics and intrinsics), and object masks. We supply image conditions from the train split for all 86 instances. ScanNet++ [37] benchmark comprises 86 instances from 6 indoor scans, each containing 10 train and test views. Instance masks are noisier than in Replica, while camera intrinsics vary per scan, posing challenges to generative models without camera awareness. Both benchmarks place objects in a cluttered scene context at metric scale, contrasting with the controlled object benchmarks evaluated in Sec. 4.2. All methods operate on per-instance mask crops, and for the depth-based SAM3D and MV-SAM3D, we additionally provide MoGe [27] depth estimates. Because object-centric pretraining frames objects in a tight field of view, whereas scene frames embed them in a wide context, VINE rescales the input intrinsics to the mask crop, aligning the effective field of view with the training distribution while preserving the true projective geometry.

Metrics. Chamfer Distance scaled by 100 (CD×100, ↓) and F-score at 0.05 m threshold (F@.05, ↑), computed after a multi-start ICP with isotropic scale alignment to the GT mesh. Severe GT mesh incompleteness on ScanNet++ biases symmetric CD, which is provided for completeness. We report paired Wilcoxon statistical significance tests [30] on both Replica and ScanNet++ instances in the supplementary.

Baselines. We compare against TRELLIS-M, ReconViaGen, SAM3D-M, and MV-SAM3D. On Replica, MV-SAM3D is evaluated both with its predicted object pose (MV-SAM3D*) and with multi-start ICP alignment (MV-SAM3D+ICP). TRELLIS-S and SAM3D-S are omitted for brevity as they perform comparably to their multidiffusion countervariants. Nonetheless, we include them in the supplementary analysis for completeness.

Results. We report combined results in Table 4. On Replica, MV-SAM3D under its predicted object pose fails to place instances at metric scale, highlighting the scale ambiguity inherent in per-view affine depth. Once the scale ambiguity is resolved, MV-SAM3D+ICP is the closest competitor: our method attains the best mean CD (4.40 vs. 4.79) and F-score (0.710 vs. 0.689), though both gaps fall within paired-test noise ($p = 0.07$ and $p = 0.21$). Against the SAM3D variants we significantly improve both CD and F-score, while ReconViaGen (F@.05 0.604) and TRELLIS-M (F@.05 0.533) trail further, the latter reflecting the base model's inaccuracy in aggregating partial captures. On ScanNet++, our method attains the best CD (3.89) and the best mean F-score: it significantly improves CD over all three depth-conditioned baselines, while the F-score margins fall within paired-test noise. Qualitative results on both real datasets are shown in Fig. 4. VINE faithfully reconstructs object shapes, whereas SAM3D multi-view adaptations resort to semantically averaged shapes. The consistent ranking across both benchmarks confirms the applicability of our approach to real indoor scenes, achieved with posed views alone despite the baselines' multi-stage pre-training and explicit depth conditioning.

4.4. Ablation Study: Two Orthogonal Design Axes

The multi-view encoder combines two separable choices: (1) *cross-view exchange* – whether per-view tokens attend to each other at all (PRoPE for SS, GTA for SLAT) before conditioning the denoiser, and (2) *aggregation* – whether

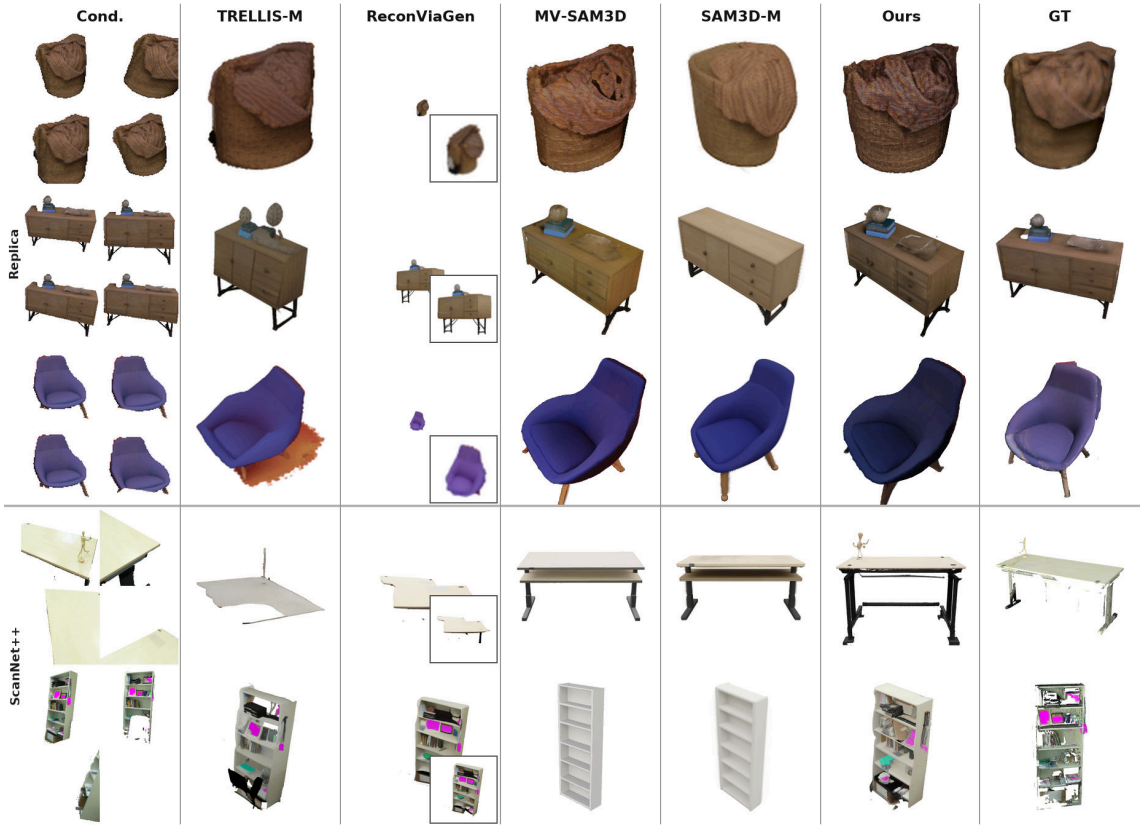


Figure 4. Qualitative comparison on Replica (top) and ScanNet++ (bottom) scene instances. On both datasets, our method recovers detailed geometry from a variable number of partial views, whereas baselines oftentimes produce incomplete or distorted reconstructions.

those tokens are then pooled into a single sequence or kept as N separate sequences. Our production model pairs each stage with a different choice on axis (2): SS pools, SLAT does not. Table 5 isolates both axes independently per stage by dropping cross-view exchange (*no encoder*) or flipping aggregation (*unpooled* for SS, *pooled* for SLAT), holding the other stage fixed to the final model.

Further, we empirically observe that dropping cross-view exchange incurs a penalty for the SS stage that grows monotonically with N, whereas flipping aggregation does not strongly affect performance on the F-metric. On the other hand, the SLAT model is not affected by removing cross-view exchange, whereas flipping the aggregation incurs a growing PSNR penalty. With this, we posit that cross-view exchange is the operative variable for geometry, while aggregation does heavy lifting for texture. Due to the page limit, we refer to the supplementary for further ablation studies.

5. Conclusion

We presented VINE, a lightweight multi-view extension of TRELLIS for object reconstruction from partial captures. A multi-view encoder attached to the frozen denoiser via LoRA aggregates an arbitrary number of posed images into geometrically consistent conditioning through camera-aware attention, enabling training on a single GPU; a curriculum view schedule bridges single-image pretraining and variable multi-view inference. On the partial-view object benchmarks VINE attains the best overall results — significantly improving texture over depth-aware SAM3D variants while matching geometry from posed RGB alone — and on the DP-Recon real scenes it achieves the best CD and F@.05 on both Replica and ScanNet++, with statistically significant CD gains over the depth-conditioned SAM3D baselines.

Stage	Variant	N=2	N=4	N=8
SS (F@.02)	Ours	**0.54**	**0.61**	**0.67**
	no encoder	0.50	0.56	0.58
	unpooled	**0.54**	<u>0.60</u>	<u>0.66</u>
SLAT (PSNR)	Ours	**18.09**	**19.05**	**19.68**
	no encoder	<u>17.97</u>	<u>18.96</u>	<u>19.59</u>
	pooled	17.49	18.27	18.79

Table 5. Geometry (SS) is sensitive to cross-view exchange (*no encoder*) but only marginally to aggregation (*unpooled*); texture (SLAT) shows the reverse pattern. **Bold**: best; <u>underline</u>: second best per column, within each stage block.

Supplementary Material – VINE

In the supplementary material, we provide details on the construction of the partial-view benchmark and compare it with alternative protocols, such as the standard full-coverage one, in Sec. 6. In Sec. 7 we categorize the training data distribution. Ablation study featuring SS and SLAT variants is provided in Sec. 8. We also investigate VINE performance given more than 8 condition views in Sec. 9. In addition to the baselines evaluated in the main paper, we report the performance of Hunyuan3D-2mv [36] and Shape-R [23] on the adapted Toys4k dataset in Sec. 10. Appearance metrics such as SSIM and LPIPS are reported in Sec. 11. Statistical significance of our results against all baselines is confirmed via paired Wilcoxon signed-rank tests in Sec. 12. We then show additional qualitative results, demonstrating our model's ability to fuse information from partial captures (Sec. 13). We also give an exact per-stage parameter count breakdown (Sec. 14), followed by runtime and memory analysis (Sec. 15). In Sec. 16, we evaluate the robustness of our model to inaccurate poses and report scene reconstruction results with a 3D open-vocabulary detector [5]. Finally, we outline limitations and future work (Sec. 17).

6. Benchmark Construction Details

Condition Views Generation. Our partial-view benchmark is based on 3D object datasets: synthetic Toys4k [24] and real OmniObject3D [31]. To simulate partial observations, we rendered each textured shape from 24 cameras whose viewing directions are drawn from a spherical Hammersley sequence, giving low-discrepancy, near-uniform coverage of the viewing sphere over the full 360° azimuthal range and a wide range of elevations. Cameras are placed at various radii (field of view ranges from 10° to 70°) to simulate close-ups and distant shots. For one fifth of the views (5 of 24), the camera looks at the object center, capturing the full object; for the remaining views, we randomly shift the look-at point off-center to simulate out-of-view, partial captures.

View-subset construction. To form a view-count set $N \in \{1, 2, 4, 8\}$, we select the subset of N views (from 24 renders on fixed elevation rings $-30^\circ/0^\circ/+30^\circ$ with 45° minimum azimuthal separation) that minimizes a combined face-level coverage score. For each candidate subset we compute three mesh-face metrics:

- *Face visibility* ($\bar{f}$): for each face, the average fraction of subset views in which its vertices are seen, averaged over all faces.
- *Face–ray incidence* ($\bar{\theta}$): mean clamped dot product between view rays and face normals over hit faces, measuring how obliquely each face is observed.
- *Pixel density* ($\bar{\rho}$): per-face image coverage (optionally normalized by projected area), scaled so the most-covered face equals 1, then averaged across faces with inverse-projected-area weights.

Candidates are generated by a complementary strategy: azimuth is split into four bins (front $|az| \le 45^\circ$, left $45^\circ < az < 135^\circ$, right $-135^\circ < az < -45^\circ$, back $|az| \ge 135^\circ$); we seed with the most forward-aligned view, add back and side views, fill the remaining slots by farthest-point sampling over view directions, and augment the pool with additional farthest-point subsets from randomized seeds. For $N = 1$, all 24 views are searched as individual candidates under the same difficulty-minimizing criterion as $N > 1$, rather than a single fixed forward-aligned candidate, so the selected view is directly comparable in difficulty to the larger-N subsets. Each metric is min–max-normalized across the candidate pool, and we retain the subset minimizing $\frac{1}{3}(\tilde{f} + \tilde{\theta} + \tilde{\rho})$.

The selected subsets serve only to define the inputs and held-out views of the benchmark. For each object and each N, the corresponding subset provides the N conditioning views given to every method, so the reconstructions, including the geometric metrics, are driven by these subsets. For appearance metrics (PSNR, LPIPS), we render each reconstruction from 10 held-out viewpoints. We report the results on difficulty-stratified slices of 200 objects: for Toys4k a *hard-200* subset (objects with the highest mean $N = 1$ Chamfer distance across methods selected from 500 random instances), and for OmniObject3D a *non-symmetric-200* subset (objects from non-symmetric categories).

Full-coverage and clustered controls. To validate that the partial-view protocol above is not interchangeable with a standard benchmark, we build two controls on the same 200 Toys4k instances. *Full-coverage* selects, for $N > 1$, evenly-spaced view rings alternating between elevation bands $\pm 35^\circ/0^\circ$, maximizing angular spread instead of the coverage-minimizing objective above. $N = 1$ uses a fixed canonical front view, shared with the clustered control below, as a common single-view anchor across both controls. *Clustered* is built to test the models' performance on redundant and less informative views. Specifically, for $N \ge 2$, views are sampled in a narrow 20° yaw cone with $\pm 6^\circ$ pitch jitter around a per-instance random anchor direction, re-observing roughly the same patch of the object instead of spreading out. Both controls use the same 10 held-out evaluation views as the proposed partial protocol, so the results are comparable across different selection strategies.

Protocol validation. Accompanying the controls in Table 3 (main), Tabs. 6 and 7 report performance for all 8 methods under the full-coverage and clustered controls (for partial-protocol numbers refer to main paper, Table 2). VINE is not the best method under either control – TRELLIS-M and ReconViaGen, in particular, benefit from full, unambiguous multi-view coverage – but it remains within the same range as the SAM3D variants and MV-SAM3D rather than trail-

Method	$N=1$ views			$N=2$ views			$N=4$ views			$N=8$ views		
	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR
Amodal3R [32]	2.78	0.588	10.01	2.84	0.565	9.66	2.50	0.601	9.78	2.25	0.650	9.81
ReconViaGen [1]	3.57	0.541	15.10	**2.66**	**0.621**	15.59	**2.02**	**0.728**	**16.92**	1.94	**0.742**	17.25
TRELLIS-S [34]	**2.69**	**0.626**	**16.03**	2.72	0.615	15.73	2.34	0.662	16.56	2.18	0.687	17.07
TRELLIS-M [34]	2.69	0.626	16.03	2.76	0.618	15.85	2.34	0.662	16.86	**1.91**	0.728	**17.79**
SAM3D-M [2]	2.98	0.578	15.69	2.78	0.591	**16.02**	2.43	0.621	16.49	2.10	0.664	16.98
SAM3D-S [2]	2.98	0.579	15.69	2.82	0.595	15.98	2.39	0.619	16.40	2.17	0.654	16.75
MV-SAM3D [10]	3.07	0.578	15.67	3.05	0.586	15.74	2.75	0.598	16.14	2.67	0.640	16.67
Ours	3.16	0.569	15.01	3.37	0.536	14.67	2.87	0.569	15.15	2.72	0.596	15.51

Table 6. Full-coverage control (Toys4k hard-200): all 8 methods, view-maximizing angular-spread conditioning. **Bold**: best; underline: second best per column. Compare against Table 1 (main, partial protocol) for the same methods.

Method	$N=1$ views			$N=2$ views			$N=4$ views			$N=8$ views		
	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR	CD×100	F@.02	PSNR
Amodal3R [32]	2.74	0.589	10.01	2.60	0.598	9.80	2.64	0.595	9.80	2.64	0.593	9.80
ReconViaGen [1]	3.58	0.540	15.10	2.65	0.622	15.83	2.67	0.621	15.85	2.69	0.614	15.74
TRELLIS-S [34]	**2.69**	**0.626**	**16.03**	**2.39**	**0.663**	**16.55**	2.50	0.654	16.43	2.44	0.658	16.47
TRELLIS-M [34]	2.69	0.626	16.03	2.44	0.657	16.44	**2.34**	**0.666**	**16.62**	**2.43**	**0.663**	**16.58**
SAM3D-M [2]	2.99	0.579	15.69	2.63	0.592	15.95	2.65	0.588	15.94	2.59	0.595	16.00
SAM3D-S [2]	2.98	0.579	15.69	2.61	0.594	15.97	2.64	0.589	15.96	2.59	0.595	16.00
MV-SAM3D [10]	3.07	0.578	15.67	3.06	0.578	15.72	3.05	0.581	15.76	2.99	0.586	15.79
Ours	3.17	0.568	15.03	3.01	0.546	14.88	2.97	0.555	14.95	2.92	0.562	14.97

Table 7. Clustered control (Toys4k hard-200): all 8 methods, narrow-cone redundant-view conditioning ($N=1$ identical to full-coverage's canonical view). **Bold**: best; underline: second best per column.

ing the field, consistent with a model designed for partial-observation robustness rather than universal generalization.

7. Training Data Composition

The SS stage draws from ObjaverseXL-Sketchfab [4] (5,000 instances), ABO [3] (4,485 instances), and HSSD [9] (6,670 instances) – a combined pool of 16,155 instances with precomputed sparse-structure latents, matching the rounded 16K-instance figure quoted in the main paper (Sec. 4.1). Figure 5 categorizes this pool by object type, assigned via keyword matching against each instance's caption (first caption variant, case-insensitive; an instance is assigned to the first matching category from a fixed 18-category keyword taxonomy, else *Other*). Furniture dominates numerically (44.7%, driven by ABO and HSSD, both indoor-object datasets), but the remaining half spans architecture, vehicles, animals, plants, kitchenware, clothing, characters, and ten further categories, with only 7.0% falling outside the fixed taxonomy entirely, clarifying the effective diversity of the training pool.

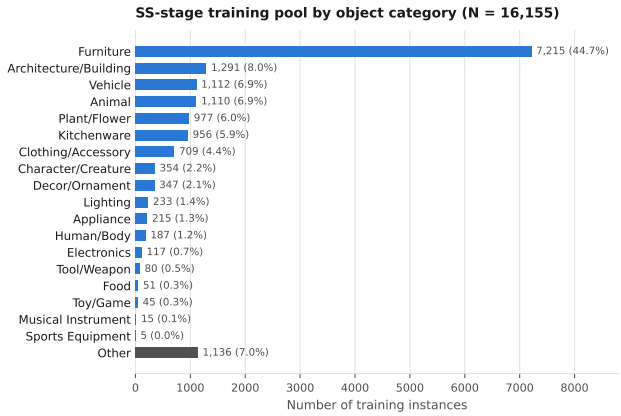


Figure 5. SS-stage training pool by object category, combined across ObjaverseXL-Sketchfab, ABO, and HSSD ($N=16{,}155$). Categories assigned via caption keyword matching; *Other* covers captions not matching the 18-category taxonomy.

8. Ablation Study

The full 17-variant ablation sweep is split across two tables: the two core design axes – cross-view exchange and aggregation, isolated independently per stage – are in the main paper (Table 5), and the remaining variants are reported here in Tab. 8. We discuss quantitative and qualitative behavior of each variant below. We separately report one further SS-stage ablation, Plücker-ray pose encoding, given its outsized and dataset-dependent effect relative to every other variant.

SS stage. Table 8 (left) reports 8 variants of our SS stage model. Full model finetuning negatively negatively affects the model performance, while LoRA on cross-attention

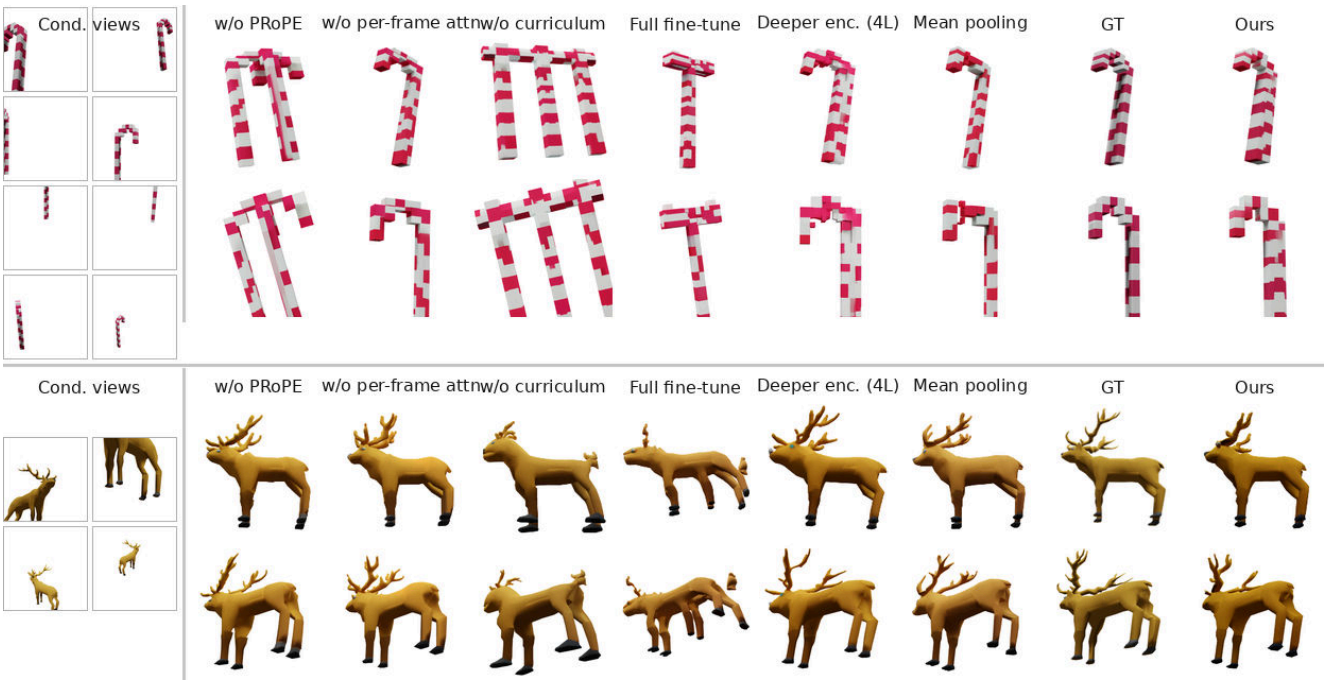


Figure 6. SS-stage ablation on two instances from the partial-view Toys4k [24] benchmark. The SLAT model (texture) is fixed across all variants. Replacing curriculum training and full fine-tuning significantly affects the geometry, while the absence of PRoPE encoding causes duplication in the sugar cane example.

SS stage — CD×100 (↓), SLAT fixed					SLAT stage — PSNR dB (↑), SS fixed				
Variant	$N=1$	$N=2$	$N=4$	$N=8$	Variant	$N=1$	$N=2$	$N=4$	$N=8$
w/o PRoPE	4.42	3.04	2.68	2.60	SS-style encoder	16.83	17.87	18.60	19.12
w/o per-frame attn.	4.27	3.16	2.50	2.29	with PRoPE	**17.01**	**18.12**	19.03	19.67
deeper encoder (4L)	4.50	3.47	2.67	2.57	with curriculum	16.95	17.92	18.96	19.53
mean pooling	**4.03**	**2.94**	2.57	3.18	LoRA r=16	16.90	17.98	18.93	19.58
multi-query pooling	4.29	3.09	**2.46**	2.29	SDXL features	16.80	17.88	18.88	19.53
LoRA (cross-attn only)	4.40	3.02	2.56	2.33					
full fine-tune (no LoRA)	5.06	4.20	3.53	3.45					
w/o curriculum	5.02	4.39	3.63	3.37					
Ours	4.42	3.11	2.59	**2.23**	Ours	16.98	18.09	**19.05**	**19.68**

Table 8. Ablation evaluation on partial-view Toys4k (200 instances), beyond the two core design axes isolated in the main paper (Table 5). **Bold**: best; underline: second best.

modules only barely makes any difference, which signals the important role these blocks carry in adaptation to the partial-view regime. Gradual introduction of additional views is shown to be very effective as random selection significantly degrades geometric quality.

Figure 6 shows all SS-stage ablations on two instances: a candy cane ($N=8$) and a reindeer ($N=4$), rendered from two different viewpoints. The SLAT stage is fixed to the final model reported in the main paper. Without PRoPE encoding, the model cannot register the input cameras and hallucinates two ghost candy canes, while the reindeer shape is less affected due to resolved ambiguity with full-view observations. Without a training curriculum, the model fails to align a single-view pretrained model towards handling multi-view conditions. Specifically, the candy cane multiplies into a fence of four copies, while the reindeer adopts a semantically wrong, dog-like shape. Full fine-tune overadapts the backbone, producing a blocky cane and a reindeer with heavily distorted body proportions. Deeper encoder (4 transformer layers), and mean pooling variants are qualitatively close to the final model on these instances, but inconsistently degrade at higher N across the benchmark as shown in Tab. 8 above.

SLAT stage. While SLAT variants perform similarly quantitatively (Tab. 8, right), qualitative comparison in Fig. 7 reveals several visual differences on a french-fries instance ($N=8$). The SS stage is fixed to the final model, so the coarse shape is identical across rows, and only the appearance varies. SS-style encoder shifts the fries from yel-

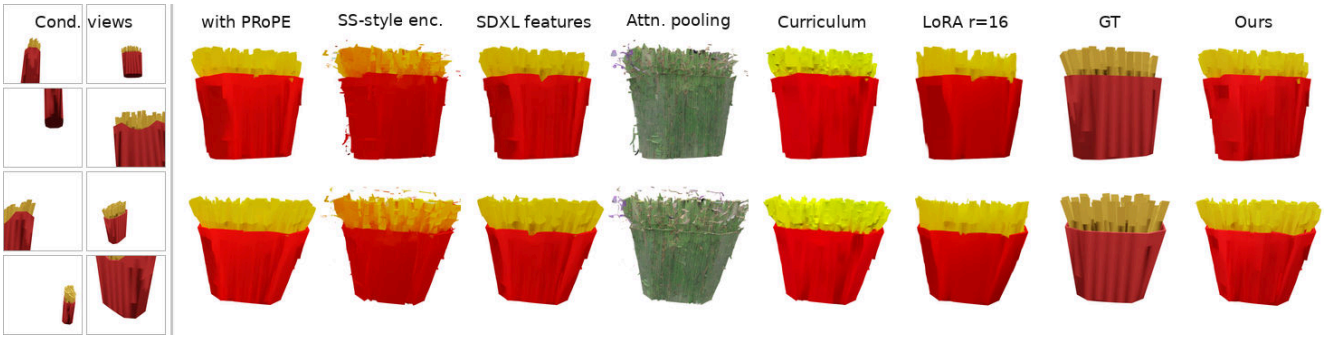


Figure 7. SLAT-stage ablation on a french-fries instance ($N=8$) from partial-view Toys4k [24] benchmark. The SS model is fixed across all variants. SS-style encoder creates geometric artifacts and turns yellow fries into orange color, while pooling completely erases color information from per-view tokens, turning reconstruction into a uniform green color.

low to orange, introducing visible surface artifacts. Attention pooling collapses per-view tokens into a single vector, erasing identity information entirely: the output is rendered as a uniform green object with no resemblance to the original red-and-yellow fries, demonstrating that pooling destroys per-view appearance information. Lower LoRA rank visibly smooths the texture and color, removing surface-induced shading on the bowl and individual fries.

Geometry/texture asymmetry: cross-view exchange axis. Table 9 reports the full CD/F@.02/PSNR numbers behind the *cross-view exchange* half of the two-axis ablation in Table 5 (main): removing the cross-view encoder refinement (2 PRoPE-conditioned transformer layers collapsed to 0, pooling itself left unchanged) from the SS stage costs an increasing penalty as N grows across all three metrics, while removing the identical component from the SLAT stage is within noise on PSNR at every N and matches Ours on CD exactly at $N=8$ (CD is only weakly affected by SLAT changes to begin with, since geometry is fixed by the SS stage). This is consistent with the qualitative pattern above: geometry synthesis needs the per-view tokens to exchange pose-aware context with each other before being pooled, since resolving shape from multiple partial observations is inherently a cross-view reasoning problem; texture synthesis does not, since each view already locally constrains its own appearance and gains little from attending to the others. The complementary *aggregation* axis (unpooled SS, pooled SLAT) shows the reverse sensitivity pattern and is reported in Table 5 (main).

Single-view LoRA baseline: adaptation vs. fusion. To separate the gain from partial-data *adaptation* from the gain from multi-view *fusion*, we train the baseline isolating the former: the frozen TRELLIS denoiser adapted with LoRA on the *same* 16K partial-view instances, but with single-image conditioning and no multi-view encoder, so it is architecturally limited to $N=1$. Table 10 compares it to VINE. At $N=1$, where no fusion is possible, this single-view specialist is modestly ahead on all three metrics (CD$\times$100 3.94 vs. 4.42, F@.02 0.461 vs. 0.429, PSNR 17.41 vs. 16.98), confirming that partial-data adaptation alone already recovers most of VINE's absolute quality and accounts for the strong $N=1$ numbers relative to the tuning-free baselines in Table 1 (main). VINE surpasses it on every metric once a second view is available and widens the margin through $N=8$, isolating the multi-view encoder's contribution as *fusion* of complementary evidence rather than mere exposure to partial-view training data.

We further isolate the encoder's contribution from the aggregation *mechanism* itself: since the single-view specialist above has no way to consume $N>1$ views at all, a skeptic could argue VINE's edge at $N\geq 2$ merely reflects having *any* multi-view mechanism, not a good one. We test this directly by applying TRELLIS-M's own naive multidiffusion averaging to the partial-data-adapted TRELLIS + LoRA weights. *TRELLIS+LoRA (naive-mv)* in Tab. 10 beats TRELLIS-M outright at every N (e.g. CD$\times$100 2.43 vs. 4.53 at $N=8$), confirming that partial-data adaptation alone already explains most of the gap to TRELLIS-M. Yet VINE still beats this naive-averaging baseline at every $N\geq 2$ on all three metrics. It is worth noting that naive averaging is a strong baseline on this object-centric benchmark, so the residual margin is rather modest (0.2–0.6 CD$\times$100), yet it is consistent across all metrics and view counts. This isolates the multi-view encoder's contribution as genuine *learned* fusion, not merely the presence of an aggregation mechanism.

Pose encoding: a Toys4k win that does not transfer to real scenes. We separately report one SS-stage ablation, additive per-view Plücker-ray token embeddings in place of PRoPE's relative pose-conditioned attention bias, because its magnitude and direction differ from every other variant in Tab. 8. On partial-view Toys4k it is the single largest effect in the entire sweep: SS CD drops by 34–41% relative to

Variant	N=1 CD↓	F↑	PSNR↑	N=2 CD↓	F↑	PSNR↑	N=4 CD↓	F↑	PSNR↑	N=8 CD↓	F↑	PSNR↑
Ours	4.42	0.429	16.98	3.11	0.541	18.09	2.59	0.613	19.05	2.23	0.670	19.68
SS, no encoder	4.34	0.426	17.01	3.40	0.498	17.86	2.93	0.559	18.61	2.65	0.584	18.99
SLAT, no encoder	4.36	0.433	16.86	3.06	0.541	17.97	2.52	0.617	18.96	2.23	0.670	19.59

Table 9. Full metrics for the cross-view exchange axis of the geometry/texture asymmetry ablation (200-instance partial-view Toys4k; see Table 5 (main) for the aggregation axis). CD×100 (↓), F@.02 (↑), PSNR dB (↑). SS-side degradation (CD, F, and PSNR) widens monotonically with N; the SLAT-side variant is statistically indistinguishable from Ours on PSNR at every N.

Metric	Method	N=1	N=2	N=4	N=8
CD×100 ↓	TRELLIS-M [34]	6.47	5.59	4.98	4.53
	+LoRA (naive-mv)	**3.94**	3.73	2.87	2.43
	Ours	4.42	**3.11**	**2.59**	**2.23**
F@.02 ↑	TRELLIS-M [34]	0.351	0.395	0.444	0.478
	+LoRA (naive-mv)	**0.461**	0.518	0.610	0.642
	Ours	0.429	**0.541**	**0.613**	**0.670**
PSNR ↑	TRELLIS-M [34]	16.01	16.98	17.79	18.35
	+LoRA (naive-mv)	**17.41**	17.64	18.98	19.59
	Ours	16.98	**18.09**	**19.05**	**19.68**

Table 10. Single-view LoRA baseline (TRELLIS-M+LoRA fine-tuned on the same 16K partial-view instances) vs. VINE on the partial-view Toys4k hard-200 benchmark. **Bold**: best per column; underline: second best. The single-view specialist wins only at N=1 (no fusion possible); VINE wins every $N \geq 2$ over both the naive-mv baseline and TRELLIS-M, isolating the multi-view encoder's contribution as learned, not merely present, fusion.

Toys4k, SS CD×100 (↓)	N=1	N=2	N=4	N=8
Ours (PRoPE)	4.42	3.11	2.59	2.23
Plücker rays	**2.64**	**1.86**	**1.54**	**1.48**
Replica	CD×100 (↓)		F@.05 (↑)	
Ours (PRoPE)	**4.40**		**0.710**	
Plücker rays	9.30		0.543	
ScanNet++	CD×100 (↓)		F@.05 (↑)	
Ours (PRoPE)	**3.89**		**0.695**	
Plücker rays	8.23		0.466	

Table 11. Plücker-ray pose encoding vs. PRoPE on three datasets. Despite achieving impressive results on object-centric Toys4k, Plücker-ray-based variant degrades significantly on real-scene captures.

Ours at every N (Tab. 11), while the identical swap barely moves SLAT PSNR (± 0.06 dB), consistent with the geometry/texture asymmetry above. This result is large enough on its own to warrant checking beyond a single benchmark: on Replica, the same SS-stage swap more than doubles CD (9.30 vs. 4.40) and drops the F-score by 0.17 (0.543 vs. 0.710). The same regression reproduces on a second, independent real-scene benchmark, ScanNet++: CD more than doubles (8.23 vs. 3.89) and F-score drops by 0.23 (0.466 vs. 0.695).

Crucially, this real-scene deficit does not close with more views: sweeping the conditioning views over $N \in 1, 2, 4, 8$ on Replica, PRoPE improves monotonically (5.26 → 4.40 CD×100, 0.66 → 0.71 F@.05), whereas the Plücker variant stays flat (∼9.1–9.6 CD×100, ∼0.54 F@.05), already trailing at N=1 where no fusion is possible and unable to exploit the additional views. Since Plücker instead improves with N on Toys4k, its absolute per-view rays support cross-view fusion only for the centered object-capture rig it was trained on.

With that, we conclude that Plücker's absolute per-view ray encoding is well-suited for an object-centric capture but does not transfer to real scene captures, whereas PRoPE's relative formulation is robust across all three benchmarks.

9. VINE Beyond $N = 8$ Views

VINE conditions on an arbitrary number of posed views, so we test its fusion behavior past the N=8 evaluation setting. Table 12 extends conditioning to $N \in \{10, 12, 14\}$ on OmniObject3D (200 instances), scoring every N against the same fixed, disjoint set of 10 held-out views. VINE peaks at N=8 on CD, PSNR, and F@.02 simultaneously and only mildly declines thereafter (CD×100 2.30 → 2.42, PSNR 20.31 → 20.18 at N=14): it neither collapses nor keeps improving once the added views become redundant. The strongest multi-view baseline, SAM3D-M, saturates earlier and flatter (CD×100 ≈2.7–2.8 from N=4 on). In comparison, VINE stays ahead at every N up to 14, though the CD margin narrows as both approach the information ceiling of the view set.

Method	Metric	N=1	N=2	N=4	N=8	N=10	N=12	N=14
Ours	CD×100 ↓	3.79	3.10	2.57	**2.30**	2.37	2.46	2.42
	F@.02 ↑	0.485	0.547	0.601	**0.626**	0.614	0.610	0.608
	PSNR ↑	18.15	18.86	19.90	**20.31**	20.25	20.12	20.18
SAM3D-M	CD×100 ↓	4.13	3.26	2.78	2.79	2.81	2.82	2.72
	F@.02 ↑	0.464	0.517	0.587	0.585	0.578	0.582	0.589
	PSNR ↑	17.88	18.73	19.48	19.64	19.64	19.65	19.72

Table 12. VINE-SAM3D-M behavior beyond the N=8 evaluation maximum on partial-view OmniObject3D (200 instances). VINE peaks at N=8 (**bold**) on all three metrics and plateaus mildly through N=14, staying ahead of SAM3D-M at every N.

10. Hunyuan3D-2mv and Shape-R Evaluation

Hunyuan3D-2mv [36] and Shape-R [23] are conditioned differently from every baseline in Table 1 (main). Specif-

ically, Hunyuan3D-2mv accepts a fixed set of canonical views ("front" / "left" / "back" / "right") and supports at most $N{=}4$. With that, we steer our evaluation to its own best-case selection of these views rather than adhering to our partial-view benchmark. Shape-R model expects a multimodal condition, with notable differences to our benchmark, as it requires a text prompt and a 3D point cloud in addition to posed images. Following the model's official pipeline, we obtain a 3D point cloud via MapAnything [8] using the same conditioning views, given GT camera poses. Both baselines are geometry-only and do not produce a textured output. All numbers below use dense-mesh GT and the same single-start ICP-alignment protocol as Table 1 (main). Nonetheless, the tables should not be explicitly compared and used for reference, instead.

Swapping MapAnything's reconstructed point cloud for the GT point cloud raises Shape-R's F@.02 by roughly $4\times$ (e.g. $0.113 \rightarrow 0.458$ at $N{=}4$) and, notably, makes it stable across view count. Moreover, MapAnything-conditioned Shape-R degrades as N grows ($0.144 \rightarrow 0.113$ from $N{=}1$ to $N{=}4$), while GT-point Shape-R stays flat, as point-cloud condition dominates the outcome once it is clean. This indicates that on this benchmark, MapAnything's reconstruction from sparse partial views is the primary bottleneck for Shape-R performance.

11. Additional Appearance Metrics

Beyond the PSNR reported in Tables 1 and 2 (main), we additionally report SSIM [28] and LPIPS [38] on the same held-out viewpoints in Tab. 14. The pattern is consistent with the main-paper F-score results: the depth-aware SAM3D variants – which have direct access to metric depth at test time – hold a small but consistent edge over VINE and the other RGB-only methods on both SSIM and LPIPS at most view counts. VINE's LPIPS gap to the closest SAM3D variant narrows monotonically with more views on both benchmarks, nearly closing by $N = 8$ on Toys4k (0.088 vs. SAM3D-M's 0.086) and reversing on OmniObject3D, where VINE attains the best LPIPS from $N = 8$ (0.091 vs. 0.094–0.095 for the SAM3D variants). SSIM is comparatively flat across methods and view counts (excluding Amodal3R, which trails throughout, all methods lie within 0.02–0.03 of one another).

12. Statistical Significance

We run one-sided paired Wilcoxon [30] signed-rank tests (H_1: "VINE better than a baseline") to establish the statistical significance of the evaluations in the main paper. The test results are reported in Tab. 15, Tab. 16, Tab. 17, and Tab. 18. For CD$\times$100, negative Δ means VINE is better; for F-score and PSNR, positive Δ means VINE is better. *wins*/n counts instances where VINE outperforms the baseline; $r = Z/\sqrt{n}$ is the effect size via the normal approximation, where Z is the normal-approximation z-statistic of the Wilcoxon signed-rank test and n corresponds to the number of paired instances.

Scene benchmarks. Tables 15 and 16 cover the Replica [25] and ScanNet++ [37] scenes from the DP-Recon benchmark [21] (86 instances each, F-score radius $r = 0.05\,\text{m}$), evaluated with the multi-ICP alignment and isotropic re-scaling as per protocol in the main paper. VINE significantly outperforms SAM3D-M and SAM3D-S on Replica (both CD and F-score, $p < 0.05$). Against MV-SAM3D on Replica and all baselines on ScanNet++ the differences are directionally positive but do not reach significance at $n = 86$, consistent with the small absolute margins (≤ 0.02 on F-score) reported in Table 3 of the main paper.

Partial-view benchmarks. Tables 17 and 18 cover partial-view Toys4k [24] and OmniObject3D [31] benchmarks, evaluated at view counts $N \in \{1, 2, 4, 8\}$. At $N = 1$ the differences are generally not significant, consistent with the single-view geometry plateau discussed in Sec. 17. On OmniObject3D, VINE is significantly better on most metrics against all three baselines from $N = 2$ onward, most consistently on PSNR (significant at every $N \geq 1$ for all baselines). On Toys4k, against these same closest SAM3D/MV-SAM3D baselines, significance is sparser and concentrated at $N = 8$; PSNR and F@.02 differences reach significance for SAM3D-M and SAM3D-S at $N = 8$, while CD and the MV-SAM3D comparisons remain largely within paired-test noise across N.

13. Additional Qualitative Results

View-count progression. Figure 8 shows how reconstruction quality evolves as the number of input views grows from $N = 1$ to $N = 8$. TRELLIS-S and TRELLIS-M struggle to combine information from additional views as geometry collapses for $N \in \{2, 4\}$. While MV-SAM3D shows consistent geometry on the locomotive example, the deer reconstruction exhibits incorrect fusion from partial observations with additional limbs for $N \in \{2, 4\}$. Amodal3R produces distorted geometry across all view counts, while ReconViaGen incorrectly scales the shapes due to the VGGT pose bias. In contrast, our model steadily accumulates geometric and appearance details with each additional view, producing compelling partial-view reconstructions.

Geometry and texture comparison. Figure 10 supplements the main paper with further instances spanning view counts $N \in \{1, 2, 4, 8\}$ on Toys4k [24] and OmniObject3D[31] benchmarks. Amodal3R [32] fails to estimate correct geometry and texture for examples such as the chess pieces (rows 3 and 7) and the earplug (row

Method	N=1		N=2		N=4	
	CD×100	F@.02	CD×100	F@.02	CD×100	F@.02
Hunyuan3D-2mv [36]	8.14	0.145	7.43	0.157	7.04	0.177
Shape-R [23]	8.66	0.144	10.29	0.112	10.48	0.113
Shape-R (GT)	3.27	**0.462**	3.24	**0.469**	3.40	**0.458**
Ours	4.42	0.429	3.11	0.541	2.59	0.613

Table 13. Hunyuan3D-2mv on the canonical-view and Shape-R on the partial-view Toys4k [24] (200 instances). **Bold**: best per column among the two Shape-R variants. *Shape-R (GT)* uses a 3D point cloud sampled from a GT mesh surface, compared to the standard MapAnything-based condition. VINE (Ours) is provided for reference.

Method	Toys4k (hard-200)								OmniObject3D (non-symmetric-200)							
	SSIM (↑)				LPIPS (↓)				SSIM (↑)				LPIPS (↓)			
	N=1	N=2	N=4	N=8	N=1	N=2	N=4	N=8	N=1	N=2	N=4	N=8	N=1	N=2	N=4	N=8
Amodal3R	0.842	0.842	0.847	0.846	0.203	0.202	0.199	0.199	0.850	0.844	0.845	0.846	0.198	0.204	0.203	0.202
ReconViaGen	0.898	0.901	0.904	0.903	0.119	0.112	0.112	0.114	0.901	0.901	0.901	0.901	0.117	0.115	0.117	0.117
TRELLIS-S	0.891	0.898	0.900	0.902	0.125	0.114	0.108	0.106	0.893	0.897	0.902	0.903	0.120	0.114	0.106	0.106
TRELLIS-M	0.891	0.897	0.905	0.907	0.125	0.114	0.104	0.098	0.893	0.897	0.904	0.905	0.120	0.112	0.105	0.103
SAM3D-M	**0.905**	0.910	**0.916**	**0.918**	**0.113**	0.102	**0.089**	**0.086**	**0.904**	0.907	**0.910**	**0.911**	**0.110**	0.104	**0.095**	0.095
SAM3D-S	**0.905**	**0.911**	0.915	0.917	**0.113**	**0.101**	**0.089**	0.088	**0.904**	**0.908**	**0.910**	**0.911**	**0.110**	**0.103**	**0.095**	0.094
MV-SAM3D	0.904	0.909	0.915	0.917	0.115	0.103	**0.089**	0.087	0.902	0.907	**0.910**	0.910	0.111	**0.103**	**0.095**	0.095
Ours	0.892	0.899	0.906	0.911	0.123	0.106	0.095	0.088	0.894	0.898	0.904	0.906	0.111	0.105	**0.095**	**0.091**

Table 14. SSIM (↑) and LPIPS (↓) on the partial-view Toys4k [24] (hard-200) and OmniObject3D [31] (non-symmetric-200) benchmarks, evaluated on the same held-out viewpoints as the main paper. **Bold**: best; underline: second best per column, within each benchmark; ties (at the reported precision) share the mark.

Baseline	**Metric**	Δ	**wins**/n	r	p
MV-SAM3D	CD×100 (↓)	−0.39	46/86	+0.16	0.071
	F@.05 (↑)	+0.020	44/86	+0.09	0.206
SAM3D-M	CD×100 (↓)	−0.83	51/86	+0.29	**0.004**
	F@.05 (↑)	+0.058	48/86	+0.27	**0.006**
SAM3D-S	CD×100 (↓)	−0.68	50/86	+0.25	**0.011**
	F@.05 (↑)	+0.049	44/86	+0.20	**0.033**

Table 15. Wilcoxon paired test on 86 Replica instances [21] $\Delta = \overline{\text{VINE}} - \overline{\text{baseline}}$; $\overline{\text{VINE}}$ (Table 3, main): CD×100 = 4.40, F@.05 = 0.710. **Bold**: statistical significance, $p < 0.05$.

Baseline	**Metric**	Δ	**wins**/n	r	p
MV-SAM3D	CD×100 (↓)	−0.80	57/86	+0.39	<**0.001**
	F@.05 (↑)	+0.024	51/86	+0.12	0.134
SAM3D-M	CD×100 (↓)	−0.72	56/86	+0.35	<**0.001**
	F@.05 (↑)	+0.018	51/86	+0.10	0.181
SAM3D-S	CD×100 (↓)	−0.77	59/86	+0.38	<**0.001**
	F@.05 (↑)	+0.014	51/86	+0.12	0.138

Table 16. Wilcoxon paired test on 86 ScanNet++ instances. $\Delta = \overline{\text{VINE}} - \overline{\text{baseline}}$; $\overline{\text{VINE}}$ (Table 3, main): CD×100 = 3.89, F@.05 = 0.695. **Bold**: statistical significance, $p < 0.05$.

6). ReconViaGen [1] scale differs from GT as it relies on the VGGT poses [26] implicitly. SAM3D [2] and MV-SAM3D [10] show competitive performance to our VINE due to their explicit 3D reliance on pointmaps [27]. TRELLIS-S [34] struggles with fusing partial-view information, especially from a high number of condition views. In Fig. 9 we further show a few more examples of ScanNet++ [37] reconstruction from real captures. Our model reconstructs objects from partial observations, preserving their original details and faithfully completing them.

Scene Layout Estimation. As described in the real-scene evaluations, we fit generated meshes into the metric GT coordinate system using multi-start ICP with isotropic rescaling. Rendering of scans 3, 5 and 6 with the corresponding GT scene meshes is shown in Fig. 11. We investigate scene reconstruction accuracy using existing object pose regression methods in Sec. 16.

14. Parameter Count Breakdown

Table 19 gives the exact, per-stage parameter count. We fine-tune TRELLIS-image-large model, consisting of the SS and SLAT stages (0.56B and 0.60B each) and accounting for a total of 1.2B parameters. The multi-view encoder, which we introduce for multi-view adaptation, is not a single fixed size: the SS encoder contains per-frame self-attention and an attention-pooling head, giving it 37.8M parameters, while the SLAT encoder consists only of the GTA cross-view layers (12.6M), since SLAT keeps per-view tokens unpooled. LoRA adapters target the query/key/value and output projections of both self- and cross-attention in all 24 blocks of each denoiser, at rank 16 (SS) and 64 (SLAT). With that, LoRA alone is 0.9% and 3.4% of the respective frozen denoiser, and reaches 7.2%/5.4% once the fully-

Baseline	Metric	$N=1$				$N=2$				$N=4$				$N=8$			
		Δ	wins/n	r	p	Δ	wins/n	r	p	Δ	wins/n	r	p	Δ	wins/n	r	p
SAM3D-M	CD×100 (↓)	+0.23	93/199	−0.06	0.801	−0.14	114/200	+0.14	**0.026**	+0.14	100/200	+0.01	0.449	−0.11	99/199	+0.09	0.096
	F@.02 (↑)	−0.007	93/199	−0.06	0.782	+0.028	112/200	+0.12	**0.047**	+0.011	104/200	+0.05	0.244	+0.046	111/199	+0.16	**0.013**
	PSNR (↑)	+0.02	102/199	−0.00	0.516	+0.13	104/200	+0.06	0.199	+0.11	105/200	+0.06	0.210	+0.39	108/199	+0.15	**0.016**
SAM3D-S	CD×100 (↓)	+0.20	92/199	−0.05	0.768	−0.08	109/200	+0.08	0.137	+0.11	100/200	+0.03	0.325	−0.14	106/199	+0.12	**0.044**
	F@.02 (↑)	−0.007	93/199	−0.06	0.794	+0.019	107/200	+0.07	0.177	+0.020	107/200	+0.08	0.126	+0.055	114/199	+0.20	**0.003**
	PSNR (↑)	+0.04	101/199	+0.00	0.475	+0.18	98/200	+0.03	0.317	+0.21	106/200	+0.07	0.165	+0.54	109/199	+0.20	**0.002**
MV-SAM3D	CD×100 (↓)	−0.08	94/199	−0.00	0.523	−0.37	112/200	+0.10	0.074	−0.14	96/200	−0.04	0.698	−0.54	95/199	+0.07	0.179
	F@.02 (↑)	+0.001	91/199	−0.03	0.658	+0.029	110/200	+0.11	0.056	+0.004	97/200	−0.01	0.552	+0.041	105/199	+0.10	0.071
	PSNR (↑)	+0.19	100/199	+0.06	0.195	+0.41	103/200	+0.10	0.072	+0.24	97/200	+0.04	0.298	+0.52	105/199	+0.12	**0.041**

Table 17. Wilcoxon test on partial-view Toys4k [24] benchmark ($n = 199$–200). $\overline{\text{VINE}}$ are taken from Table 1, main. $\Delta = \overline{\text{VINE}} - \overline{\text{baseline}}$; **bold**: statistical significance, $p < 0.05$.

Baseline	Metric	$N=1$				$N=2$				$N=4$				$N=8$			
		Δ	wins/n	r	p	Δ	wins/n	r	p	Δ	wins/n	r	p	Δ	wins/n	r	p
MV-SAM3D	CD×100 (↓)	−0.37	114/200	+0.14	**0.020**	−0.40	118/200	+0.18	**0.005**	−0.14	106/200	+0.07	0.159	−0.62	113/200	+0.23	**<.001**
	F@.02 (↑)	+0.020	114/200	+0.10	0.084	+0.030	119/200	+0.16	**0.012**	−0.003	103/200	+0.01	0.456	+0.041	108/200	+0.17	**0.008**
	PSNR (↑)	+0.38	116/200	+0.14	**0.021**	+0.33	118/200	+0.15	**0.016**	+0.49	115/200	+0.19	**0.003**	+1.04	132/200	+0.34	**<.001**
SAM3D-M	CD×100 (↓)	−0.32	114/200	+0.14	**0.026**	−0.23	113/200	+0.17	**0.007**	−0.16	111/200	+0.12	**0.044**	−0.51	112/200	+0.24	**<.001**
	F@.02 (↑)	+0.022	115/200	+0.11	0.064	+0.030	115/200	+0.15	**0.018**	+0.010	109/200	+0.07	0.172	+0.044	101/200	+0.18	**0.005**
	PSNR (↑)	+0.32	115/200	+0.12	**0.041**	+0.28	115/200	+0.14	**0.021**	+0.54	120/200	+0.22	**<.001**	+0.95	125/200	+0.33	**<.001**
SAM3D-S	CD×100 (↓)	−0.34	114/200	+0.14	**0.026**	−0.24	110/200	+0.15	**0.016**	−0.12	109/200	+0.10	0.089	−0.51	114/200	+0.25	**<.001**
	F@.02 (↑)	+0.022	115/200	+0.11	0.068	+0.028	112/200	+0.11	0.053	+0.004	108/200	+0.03	0.331	+0.040	109/200	+0.17	**0.007**
	PSNR (↑)	+0.31	115/200	+0.12	0.052	+0.31	117/200	+0.15	**0.020**	+0.56	118/200	+0.21	**0.001**	+0.97	125/200	+0.33	**<.001**

Table 18. Wilcoxon test on partial-view OmniObject3D [31] benchmark ($n = 200$). $\overline{\text{VINE}}$ are taken from Table 2, main. $\Delta = \overline{\text{VINE}} - \overline{\text{baseline}}$; **bold**: statistical significance, $p < 0.05$.

trainable encoder is included in the numerator.

Component	SS stage	SLAT stage
Frozen denoiser	555.5M	600.4M
Multi-view encoder (trainable)	37.8M	12.6M
LoRA adapters, r=16/64 (trainable)	5.1M	20.4M
Total trainable	**42.9M**	**33.0M**
Trainable / frozen denoiser	**7.2%**	**5.4%**

Table 19. Exact parameter counts per stage, obtained from the instantiated production models. The two denoisers together (1.16B) account for the 1.2B-parameter TRELLIS-image-large backbone [34].

15. Runtime and Memory Efficiency

We compare inference runtime and peak memory at $N = 8$ on a single NVIDIA A40. For a fair comparison, all methods consume the provided object masks (no background removal), timings exclude ICP alignment and rendering, and each method is timed on a fixed instance, 10 runs after one warm-up. At our full protocol of 30 diffusion steps per stage, VINE reconstructs in 52.6 s. The cost is dominated by denoising: the lightweight encoder and decoder add only a small fixed overhead, but the SLAT stage cross-attends to every view's token set, so the per-step cost grows with N. Accuracy, however, saturates far below 30 steps: between 5+5 and 30+30 steps per stage, F@0.02 and CD are almost unchanged, while PSNR drops by only $\sim$0.9 dB (19.49 → 18.61, Fig. 12). The fast configuration VINE (5+5 steps) runs in 10.3 s at the same geometric quality, 2.0× faster than TRELLIS multi-diffusion (20.7 s) and within 2× of TRELLIS stochastic (5.1 s), while reconstructing substantially better geometry than either (Tables 1–2, main).

Memory is where VINE stands apart among high-quality methods: it requires only 6.6 GB, versus 17.2 GB for the SAM3D variants (2.6×) at comparable quality. Table 20 and Fig. 12 summarize the accuracy–compute–memory trade-off: VINE exposes a step-count knob that trades a little texture detail for large speed-ups — the full 30+30 setting attains the best PSNR, while 5+5 matches SAM3D-S in PSNR at roughly half the time (10.3 vs 19.3 s) and a fraction of their memory. While TRELLIS-S achieves the lowest absolute runtime, VINE offers the best geometry-per-second and the lowest memory among methods of comparable accuracy.

16. Scene reconstruction with estimated poses.

In Tab. 21, we replace the oracle components of the Replica protocol with off-the-shelf estimators. Our model maintains its geometric accuracy with estimated camera calibration (both extrinsics and intrinsics) from MapAnything [8], which further validates PRoPE's robustness to inaccuracies. Replacing the multi-start GT-ICP placement with a real 6-

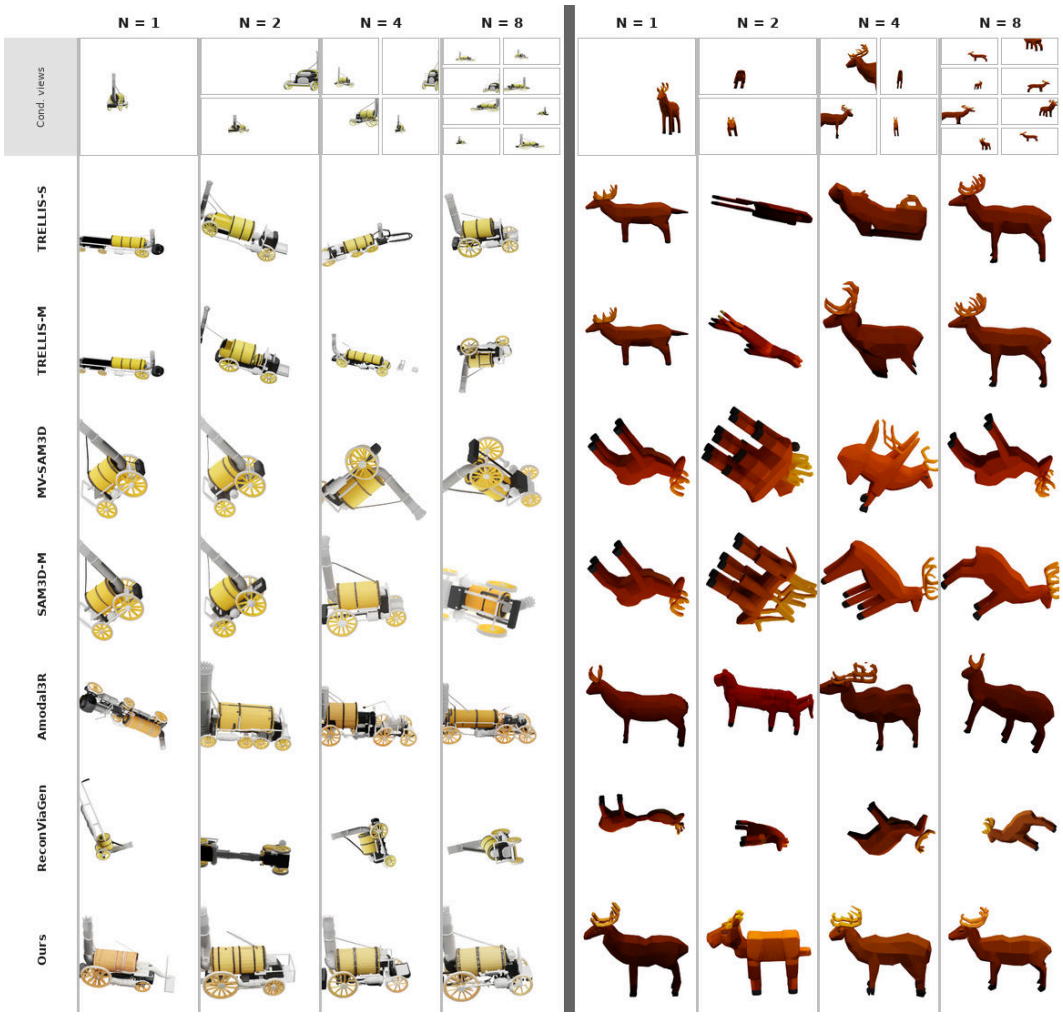


Figure 8. Qualitative results for an increasing number of input conditions on two examples from Toys4k. All baselines demonstrate incorrect deer geometry at $N \in \{2, 4\}$, struggling to fuse complementary partial-view information. In contrast, our model steadily refines geometry and appearance across all baselines, even in highly ambiguous partial-view sets.



Figure 9. Additional ScanNet++ [37] qualitative results from the DP-Recon benchmark [21]. Our model faithfully reconstructs the geometry and color of the objects.

DoF object pose from FoundationPose [29] registration or from the Boxer [5] 3D detector's oriented box places generated shapes at lower accuracy despite models being conditioned on metric-scale MoGe output [27]. Since the GT-camera mesh is identical between the GT and FoundationPose/Boxer placement rows in Tab. 21, the gap reflects the

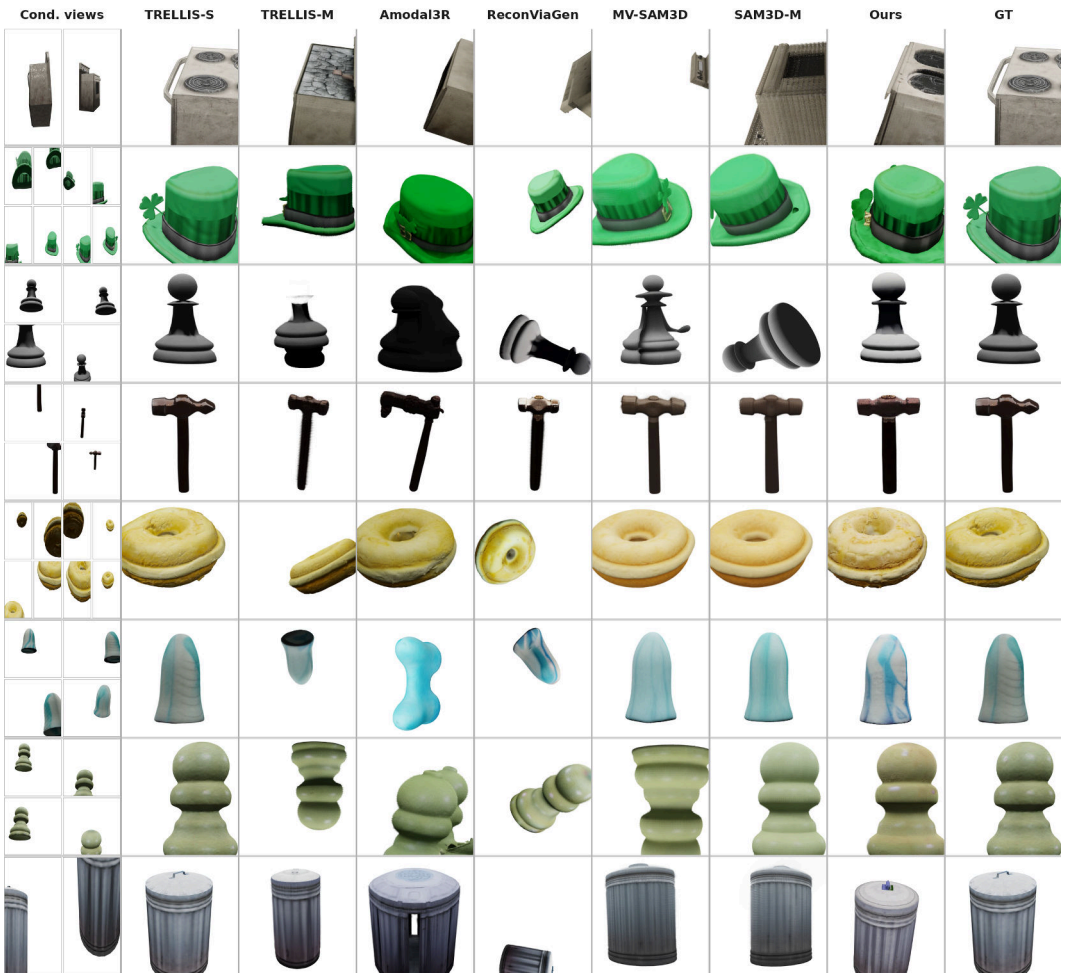


Figure 10. Additional qualitative results on the partial-view Toys4k [24] and OmniObject3D [31] benchmarks. Our model recovers the correct shape and texture in all cases, demonstrating versatility in aggregating a variable number of partial observations.

Method	PSNR ↑	Time (s) ↓	VRAM (GB) ↓
TRELLIS-S [34]	16.59	$\mathbf{5.1 \pm 0.3}$	**5.8**
TRELLIS-M [34]	16.96	20.7 ± 0.4	**5.8**
ReconViaGen [1]	14.97	11.1 ± 0.0	14.6
SAM3D-S [2]	<u>18.72</u>	19.3 ± 0.0	17.2
SAM3D-M [2]	18.67	133.9 ± 0.1	17.2
Ours (5+5 steps)	18.61	<u>10.3 ± 0.0</u>	<u>6.6</u>
Ours (30+30, full)	**19.49**	52.6 ± 0.0	<u>6.6</u>

Table 20. Per-instance inference time and peak GPU memory on OmniObject3D (200 instances, $N = 8$) reported on NVIDIA A40. Time: mean $\pm$ std over 10 runs; VRAM: peak allocated memory.

difficulty of absolute 6-DoF object localization, not of shape generation, identifying scene placement as the practical bottleneck for fully pose-free decompositional reconstruction.

Camera poses	Object placement	F@.05	CD×100
GT	multi-start ICP	**0.710**	**4.40**
MapAnything	multi-start ICP	0.703	4.41
GT	FoundationPose [29]	0.330	21.63
GT	Boxer [5] 3D OOBB	0.401	15.79

Table 21. Replica scene reconstruction with estimated camera and object poses from DP-Recon benchmark [21].

17. Limitations and Future Work

While VINE demonstrates competitive performance against depth-aware SAM3D variants, pose-free metric generation remains part of future work. Since VINE's contribution is the fusion of complementary views, its advantage emerges once multiple views are available and at $N = 1$ geometry is statistically on par with the SAM3D tuning-free baselines. Closing the geometry and texture gaps by distilling a stronger object prior, such as TRELLIS 2.0 [33] with additional material prediction, pixel-aligned Pixal3D [11] or

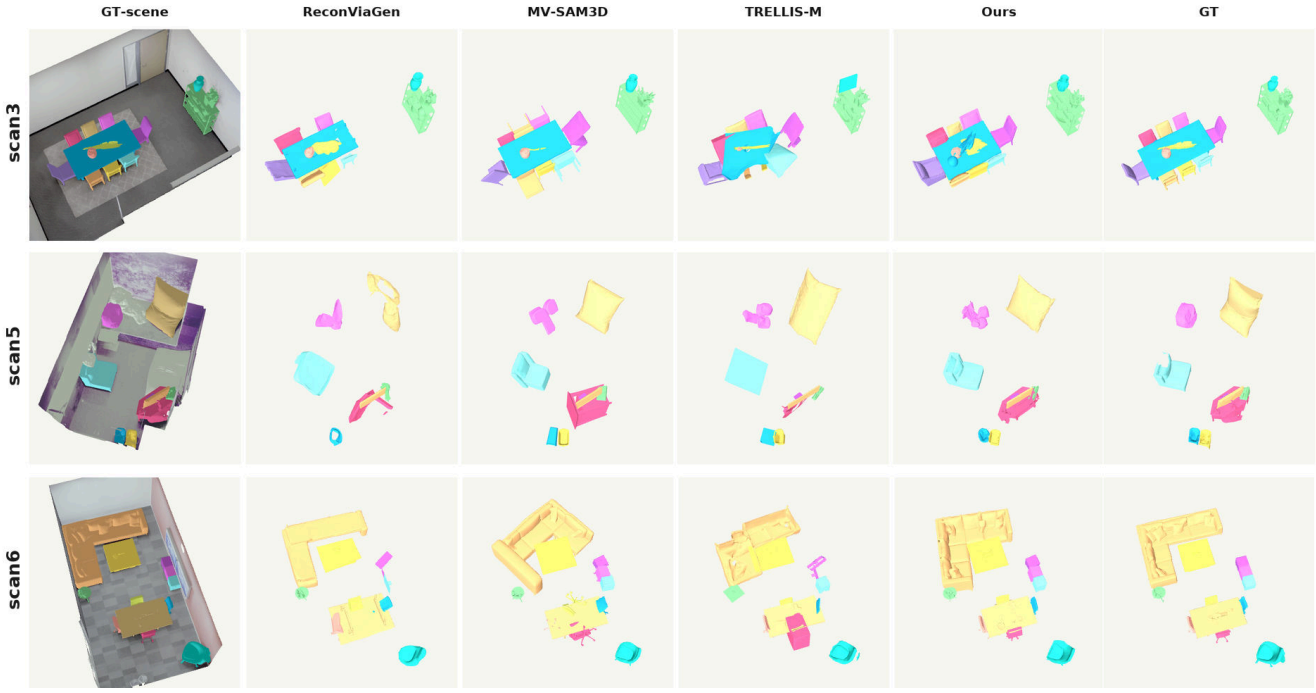


Figure 11. Scene-level reconstruction on Replica [25] scans from the DP-Recon benchmark [21] using multi-start ICP with isotropic re-scaling. Object's faithfull geometry plays a decisive role in ICP alignment success, which is reflected in the differences between each model's reconstruction.

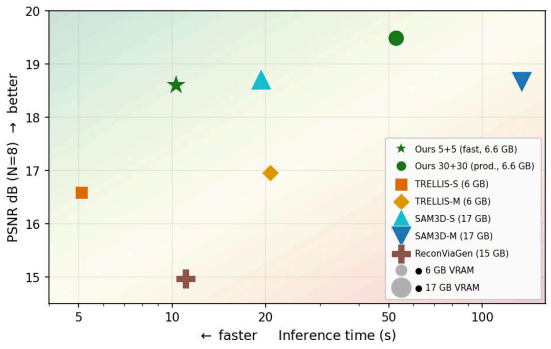


Figure 12. Efficiency–quality–memory trade-off on OmniObject3D (200 instances, $N = 8$) reported on NVIDIA A40 GPU. The time axis is log-scaled. Bubble area is proportional to peak VRAM. VINE (30+30) attains the best PSNR (19.5 dB); VINE (5+5) matches the SAM3D variants while running $2\times$ faster and using approx. $2.6\times$ less memory.

metric depth-aware SAM3D [2], is an exciting future direction.

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